

MANIK GOYAL

Backend Software Engineer | Java, Kafka, PostgreSQL, AWS

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2 years of production experience building distributed services and REST APIs on **Kafka, PostgreSQL/TimescaleDB and AWS**, supporting a video analytics platform across **400+ cameras and 200+ servers**. Scaled sustained event throughput **16x** and cut analytics export runtime by **92%**.

TECHNICAL SKILLS

Languages: Java, C# (C Sharp), C++17, SQL, JavaScript, TypeScript, HTML5, CSS3

Backend & Frameworks: Spring Boot, Hibernate/JPA, REST APIs, ASP.NET Core 8/7, Entity Framework Core, Microservices, Distributed Systems

Frontend: React.js, Angular 18, Angular Material, NPM package authoring, Component Libraries

Messaging & Streaming: Apache Kafka, Kafka Consumer Groups, RabbitMQ, Pub/Sub, WebRTC, FFmpeg

Databases: PostgreSQL, TimescaleDB, SQL Query Optimization, Indexing, Partitioning, Compression, Milvus (vector search)

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, GitHub Actions, CI/CD, Linux, Grafana, Prometheus, Observability

Practices: System Design, Clean Architecture, SOLID, Design Patterns, API Design, Performance Tuning, Code Review, Agile/Scrum, Unit Testing

Spoken: English (Professional), Hindi (Native), German (Basic)

PROFESSIONAL EXPERIENCE

Software Engineer, I2V Systems Pvt. Ltd.

Oct 2024 - Present

Gurugram, India | Kafka, PostgreSQL, TimescaleDB, AWS, REST APIs, React

- Led development of the Analytics Manager service for a **400+ camera** video analytics platform, coordinating ANPR, face recognition and intrusion events across **8+** backend and analytics services.
- Diagnosed event-processing bottlenecks and redesigned Kafka consumption around event-specific topics with dead-letter handling, lifting sustained throughput from **30 to 500+ events/sec (16x)** while isolating failed messages for replay.
- Architected reusable messaging and storage abstraction libraries spanning Kafka, local disk, NAS and AWS S3, adopted by **4+ services** and cutting new-service integration from days to hours.
- Optimized PostgreSQL and TimescaleDB ingestion with hypertables, native compression, retention policies and composite indexes, raising camera-frame processing from **70 to 300+ frames/sec**; tuned analytics database indexes to cut query latency by **35%**.
- Implemented Milvus vector similarity search for face enrollment, watchlists and deduplication, reducing vector-index memory consumption by **75%** while preserving match accuracy.
- Reengineered analytics exports over **100K+ records** with streaming reads and incremental batch processing, cutting execution time from **12 minutes to under 1 minute (92% faster)** and eliminating out-of-memory failures.
- Built bulk enrollment workflows for **5,000+ records** with server-side validation and live progress streaming; integrated government e-challan REST APIs into the ANPR violation pipeline.
- Published **10+ reusable React/NPM component libraries** consumed by web apps across a **200+ server** deployment serving 60+ concurrent users, standardizing UI behavior across products.
- Automated Docker-based release and changelog workflows with GitHub Actions, reducing release turnaround by **40%**, and instrumented production services with Grafana and Prometheus dashboards and alerts.
- Mentored **2 junior engineers** on Clean Architecture and asynchronous service patterns, and drove code reviews that cut QA-reported regressions.

PROJECTS

ParkNest, Peer-to-Peer Parking Marketplace

2025

PostgreSQL, REST APIs, React, Asynchronous

- Engineered a two-sided marketplace supporting host listings, renter bookings and prepaid credit settlement over a normalized PostgreSQL schema.
- Designed optimistic-concurrency booking safeguards and an idempotent credit ledger that prevent double-booking and duplicate transactions under overlapping requests.
- Designed REST API contracts for listing, booking and settlement workflows with server-side validation and consistent failure handling, keeping backend behavior predictable for clients.
- Applied failure-safe retry handling around booking and credit operations so repeated requests do not corrupt booking state or duplicate ledger entries.

EDUCATION

Bachelor of Technology (B.Tech), Information Technology

Dec 2021 - Jun 2025

University School of Information, Communication and Technology (USICT), GGSIP University, Delhi | **CGPA: 8.55/10**

CERTIFICATIONS & ACHIEVEMENTS

Postman API Fundamentals Certification - API design, documentation and automated testing workflows.

LeetCode: solved **800+** data structures and algorithms problems across arrays, strings, graphs, trees and algorithm design.

Adobe GenSolve: placed in the top 5 percentile nationally in the algorithmic coding competition.